KAIZEN MATHS

Lesson starter

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Newton-Raphson

Name

S1

Use the Newton-Raphson formula for one iteration.

Use one Newton-Raphson step for $f(x) = x^3 - 12$ with $x_0 = 3$.

S2

Use Newton-Raphson twice, showing each substituted value.

Use Newton-Raphson twice to solve $x^3 - 11 = 0$, starting with $x_0 = 3$.

S3

Derive the Newton-Raphson iteration formula, then apply it.

For $f(x) = x^3 - 5$, derive the Newton-Raphson iteration formula and use $x_0 = 2$ to find x_1.

S4

Use the Newton-Raphson formula for one iteration.

Use one Newton-Raphson step for $f(x) = x^3 - 3$ with $x_0 = 2$.

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For $f(x) = x^3 - 5$, derive the Newton-Raphson iteration formula and use $x_0 = 2$ to find x_1.

S4

Use the Newton-Raphson formula for one iteration.

Use one Newton-Raphson step for $f(x) = x^3 - 3$ with $x_0 = 2$.

ALG-D1B0-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Newton-Raphson

Name _____

EA1

Use Newton-Raphson twice, showing each substituted value.

Use Newton-Raphson twice to solve $x^3 - 5 = 0$, starting with $x_0 = 2$.

EA2

Derive the Newton-Raphson iteration formula, then apply it.

For $f(x) = x^3 - 2$, derive the Newton-Raphson iteration formula and use $x_0 = 2$ to find x_1.

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KAIZEN MATHS

Exit ticket

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Newton-Raphson

Name _____

EB1

Use the Newton-Raphson formula for one iteration.

Use one Newton-Raphson step for $f(x) = x^3 - 5$ with $x_0 = 2$.

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KAIZEN MATHS

Exit ticket

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Newton-Raphson

Name _____

EA1

Use Newton-Raphson twice, showing each substituted value.

Use Newton-Raphson twice to solve $x^3 - 5 = 0$, starting with $x_0 = 2$.

EA2

Derive the Newton-Raphson iteration formula, then apply it.

For $f(x) = x^3 - 2$, derive the Newton-Raphson iteration formula and use $x_0 = 2$ to find x_1.

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KAIZEN MATHS

Exit ticket

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Newton-Raphson

Name _____

EB1

Use the Newton-Raphson formula for one iteration.

Use one Newton-Raphson step for $f(x) = x^3 - 5$ with $x_0 = 2$.

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Show your working

Teacher answers & working

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S1

Use the Newton-Raphson formula for one iteration.

Use one Newton-Raphson step for $f(x) = x^3 - 12$ with $x_0 = 3$.

Answer:

$x_1 = 2.44444$.

Working:

Newton-Raphson uses $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.

$f(x) = x^3 - 12$, so $f'(x) = 3x^2$.

$f(3) = 3^3 - 12 = 15$.

$f'(3) = 3(3)^2 = 27$.

$x_1 = 3 - \frac{15}{27} = 2.44444$.

S2

Use Newton-Raphson twice, showing each substituted value.

Use Newton-Raphson twice to solve $x^3 - 11 = 0$, starting with $x_0 = 3$.

Answer:

$x_2 = 2.2376$.

Working:

$f(x) = x^3 - 11$ and $f'(x) = 3x^2$.

First iteration: f(3) = 16 and f'(3) = 27.

$x_1 = 3 - \frac{16}{27} = 2.40741$.

Second iteration: f(2.40741) = 2.9524 and f'(2.40741) = 17.38683.

$x_2 = 2.40741 - \frac{2.9524}{17.38683} = 2.2376$.

Teacher answers & working

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule · Newton-Raphson

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S3

Derive the Newton-Raphson iteration formula, then apply it.

For $f(x) = x^3 - 5$, derive the Newton-Raphson iteration formula and use $x_0 = 2$ to find x_1.

Answer:

$x_{n+1} = \frac{2x_n^3+5}{3x_n^2}$, $x_1 = 1.75$.

Working:

Start with $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.

Here $f(x_n) = x_n^3 - 5$ and $f'(x_n) = 3x_n^2$.

$x_{n+1} = x_n - \frac{x_n^3-5}{3x_n^2}$.

Write x_n as $\frac{3x_n^3}{3x_n^2}$ so the terms can be combined.

$x_{n+1} = \frac{3x_n^3-(x_n^3-5)}{3x_n^2} = \frac{2x_n^3+5}{3x_n^2}$.

$x_1 = 2 - \frac{3}{12} = 1.75$.

S4

Use the Newton-Raphson formula for one iteration.

Use one Newton-Raphson step for $f(x) = x^3 - 3$ with $x_0 = 2$.

Answer:

$x_1 = 1.58333$.

Working:

Newton-Raphson uses $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.

$f(x) = x^3 - 3$, so $f'(x) = 3x^2$.

$f(2) = 2^3 - 3 = 5$.

$f'(2) = 3(2)^2 = 12$.

$x_1 = 2 - \frac{5}{12} = 1.58333$.

Teacher answers & working

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule · Newton-Raphson

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EA1

Use Newton-Raphson twice, showing each substituted value.

Use Newton-Raphson twice to solve $x^3 - 5 = 0$, starting with $x_0 = 2$.

Answer:

$x_2 = 1.71088.$

Working:

$f(x) = x^3 - 5$ and $f'(x) = 3x^2$.

First iteration: f(2) = 3 and f'(2) = 12.

$x_1 = 2 - \frac{3}{12} = 1.75.$

Second iteration: f(1.75) = 0.35938 and f'(1.75) = 9.1875.

$x_2 = 1.75 - \frac{0.35938}{9.1875} = 1.71088.$

EA2

Derive the Newton-Raphson iteration formula, then apply it.

For $f(x) = x^3 - 2$, derive the Newton-Raphson iteration formula and use $x_0 = 2$ to find x_1.

Answer:

$x_{n+1} = \frac{2x_n^3+2}{3x_n^2}$, $x_1 = 1.5.$

Working:

Start with $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.

Here $f(x_n) = x_n^3 - 2$ and $f'(x_n) = 3x_n^2$.

$x_{n+1} = x_n - \frac{x_n^3-2}{3x_n^2}$.

Write x_n as $\frac{3x_n^3}{3x_n^2}$ so the terms can be combined.

$x_{n+1} = \frac{3x_n^3-(x_n^3-2)}{3x_n^2} = \frac{2x_n^3+2}{3x_n^2}$.

$x_1 = 2 - \frac{6}{12} = 1.5.$

Teacher answers & working

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule · Newton-Raphson

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EB1

Use the Newton-Raphson formula for one iteration.

Use one Newton-Raphson step for $f(x) = x^3 - 5$ with $x_0 = 2$.

Answer:

$x_1 = 1.75$.

Working:

Newton-Raphson uses $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.

$f(x) = x^3 - 5$, so $f'(x) = 3x^2$.

$f(2) = 2^3 - 5 = 3$.

$f'(2) = 3(2)^2 = 12$.

$x_1 = 2 - \frac{3}{12} = 1.75$.